The emulator-based Fisher for ASTRA clustering

ASTRA clustering project

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Abstract

The original ASTRA Fisher forecast had two structural weaknesses: derivatives from the HOD-mismatched ± cosmology pairs (HOD contamination – the Tier-0/1 subtraction saga), and a covariance $C = C_{\rm CV}$ that implicitly assumes a perfect emulator (which we showed over-states the ASTRA value-add). The Tier-3 emulator removes both: it gives *HOD-clean* derivatives $\partial\xi/\partial\theta$ at fixed HOD for the full w_0w_aCDM+ parameter set (including w_0, w_a), and we have *measured* the emulator-error covariance $C_{\rm emu}$. This note defines the emulator-based Fisher ($C = C_{\rm CV} + C_{\rm emu}$, joint cosmology+HOD with HOD marginalisation), validates it against the MCMC, and uses it to forecast the campaign payoff. It is the companion to `mlp_emulator_note` (which builds the emulator and the inference prototype).

1 Why a new Fisher

The Fisher matrix is $F_{ab} = \sum(\partial_a\xi)\, C^{-1}\,(\partial_b\xi)$; its quality is set entirely by the derivatives and the covariance. The original forecast (`notes/fisher`, `vector_search`) had known problems in both:

- **Derivatives.** Central differences over the ±2–3% AbacusSummit pairs are *not* HOD-matched, so the HOD mismatch contaminated $\partial\xi/\partial\theta$ at a level comparable to the cosmology signal — worst for σ_8 (the raw derivative was ∼11% of the expected 2ξ). Tiers 0/1 fought this with a noise model and an explicit $\partial\xi/\partial\theta_{\rm HOD}$ subtraction. And the ±-pair grid only gave the **4 ΛCDM** parameters.
- **Covariance.** Using $C = C_{\rm CV}$ (subbox) assumes the model is exact. The value-add test (`mlp_emulator_note` § 13) proved this is optimistic: in real inference $C = C_{\rm CV} + C_{\rm emu}$, and the environment-leg $C_{\rm emu}$ down-weights exactly the legs the $C_{\rm CV}$-only Fisher crowned. The Fisher leg-ranking and FoM were therefore not predictive of real inference.

2 Method

The trained, CV-weighted emulator $\xi(\theta_{\rm cosmo}, \theta_{\rm HOD})$ repairs both:

1. **HOD-clean derivatives.** $\partial\xi/\partial\theta$ by central finite-difference of the emulator at the $c000$ fiducial, varying one parameter at a time with *everything else (including the 12 HOD) held fixed.* This is HOD-clean by construction — no mismatch, no subtraction — and covers all **8** w_0w_aCDM+ parameters $(\omega_b, \omega_{\rm cdm}, h, n_s, \alpha_s, N_{\rm ur}, w_0, w_a)$ plus the 12 HOD gradients, from one model.
2. **Realistic covariance.** $C = C_{\rm CV} + C_{\rm emu}$, with $C_{\rm CV}$ the pooled-subbox cosmic variance (box volume) and $C_{\rm emu}$ the emulator-error covariance from a CV-weighted leave-one-cosmology-out (the same residuals that set the inference error budget).

3. **Joint cosmology+HOD Fisher.** $F = D^{\top} C^{-1} D$ over the 8 cosmology + 12 HOD derivatives. *Conditional* errors fix the HOD (invert the cosmology block); *marginalised* errors add a Gaussian yuan23 HOD prior (width = the training-set HOD spread) and invert the full matrix, taking the cosmology block.

4. **Validation.** Compare the Fisher σ to the curated-vector MCMC posteriors (same vector, parameters, HOD treatment). Agreement bounds where the Gaussian-linear Fisher is trustworthy.

5. **Campaign forecast.** Scale $C_{\rm emu} \to \alpha\, C_{\rm emu}$ and trace the FoM vs α — the perfect-emulator ($\alpha \to 0$) limit is what the campaign approaches by driving $C_{\rm emu}$ down, quantifying its payoff.

3 Scripts

A single GPU build, then fast CPU analysis (all in `scripts/`):

- `emufisher_lib.py` — shared: leg loading, subbox $C_{\rm CV}$, emulator derivatives, the joint Fisher with HOD marginalisation, corner-ellipse plotter.
- `emufisher_build.py` — the GPU step: CV-weighted emulator → derivatives D (270-D vector × 20 params), $C_{\rm CV}$, $C_{\rm emu}$ (LOCO) → `data/emulator_tier3/emufisher_build.npz`.
- `emufisher_forecast.py` — marginalised σ and FoM per data vector (full 2PCF / curated / full+ASTRA); corner ellipses.
- `emufisher_campaign.py` — FoM vs $C_{\rm emu}$ scaling α.
- `emufisher_validate.py` — Fisher vs. the curated MCMC σ.

The candidate vector is the full 2PCF + the eight environment stems, monopole and quadrupole (18 legs); sub-vectors are column masks (no re-build).

4 Results

The build (`emufisher_build.npz`) provides the derivative matrix D (a 270-D vector × 20 parameters), $C_{\rm CV}$, and $C_{\rm emu}$ from the CV-weighted LOCO; the environment-leg emulator error is sizeable, median $\sqrt{\operatorname{diag} C_{\rm emu}/C_{\rm CV}} = 7.4$ over the 18-leg vector.

4.1 Forecast over data vectors

Table **??** gives the HOD-marginalised cosmology σ and the FoM (relative to the full-2PCF baseline) for three vectors: the plain 2PCF (monopole+ quadrupole), the greedy-*curated* vector (2PCF + the void-random-cross monopole), and *full+all ASTRA* (the full 2PCF + the eight extreme void/knot environment stems, $\ell = 0, 2$). The corner ellipses are in Fig. **??**.

The decisive point is that the **FoM gains far exceed the per-parameter σ gains**: the curated vector tightens each marginal error only ×1.1–1.4 yet multiplies the FoM by 14×, and full+ASTRA gives ×1.8–2.2 per parameter but 150× in FoM. Since the FoM is the inverse *volume* of the joint error ellipsoid, this gap is **degeneracy-breaking**: the environment legs rotate and de-correlate the cosmology ellipse far more than they shrink any single axis — exactly the ASTRA thesis, now visible in the emulator-based forecast.

vector	n_b	$\sigma(\omega_b)$	$\sigma(\omega_{\rm cdm})$	$\sigma(h)$	$\sigma(n_s)$	$\sigma(w_0)$	$\sigma(w_a)$
full 2PCF	30	0.0049	0.010	0.050	0.059	0.14	0.57
curated	45	0.0044	0.0075	0.037	0.041	0.099	0.48
full + all ASTRA	270	0.0022	0.0056	0.027	0.029	0.068	0.25

Table 1: Marginalised σ (HOD-marginalised, $C = C_{\rm CV} + C_{\rm emu}$). FoM relative to the full 2PCF: curated **14.3**×, full+ASTRA **150**×. Per-parameter σ ratios vs the 2PCF: curated ×1.1–1.4, full+ASTRA ×1.8–2.2.

4.2 Validation against the MCMC

On the curated monopole vector with the 4 ΛCDM parameters, the Fisher σ are 2.3–5.9× **looser than the MCMC** posteriors (ratios: ω_b 5.9, $\omega_{\rm cdm}$ 3.5, h 5.8, n_s 2.3, HOD-marginalised; HOD-fixed is similar). The Fisher is therefore *conservative* in absolute terms here, while its *relative* predictions (the value-add ratios) match the MCMC value-add (×1.2–1.7). The absolute gap is consistent with a finite-difference derivative step that under-estimates the local gradient and with the MCMC posterior being sharpened by the bounded prior and the specific mock; it should be refined before quoting absolute σ, but does not affect the leg-ranking or FoM *comparisons*.

4.3 Campaign forecast: FoM vs emulator error

Scaling $C_{\rm emu} \to \alpha C_{\rm emu}$ traces what driving the emulator error down buys (Fig. **??**). The FoM is **steeply gated** by $C_{\rm emu}$:

α	1.0	0.5	0.3	0.1	0.03	0.01	$\to 0$
median $C_{\rm emu}/C_{\rm CV}$	7.4	5.3	4.1	2.4	1.3	0.74	0
FoM / FoM(α=1)	1	4.7	13	97	610	2341	10^5

Bringing the median environment-leg error to $C_{\rm emu} \approx C_{\rm CV}$ ($\alpha \sim 0.03$) already unlocks ∼600× FoM over the current emulator. This is the quantitative campaign target: the value-add is realised only as $C_{\rm emu}$ falls below cosmic variance, which (per the emulator note) is set by cosmology coverage.

5 Caveats

The covariance is still the subbox estimate (no independent-mock suite under the no-new-HOD constraint). Derivatives are at one fiducial (c000); the linear-Gaussian Fisher is bounded by the MCMC validation (§ 5). $C_{\rm emu}$ is from a single-phase LOCO and is cosmology-averaged; a neighbourhood-matched $C_{\rm emu}$ would refine the broad-prior numbers. w_0, w_a rely on the quadrupole and remain the weakest directions.

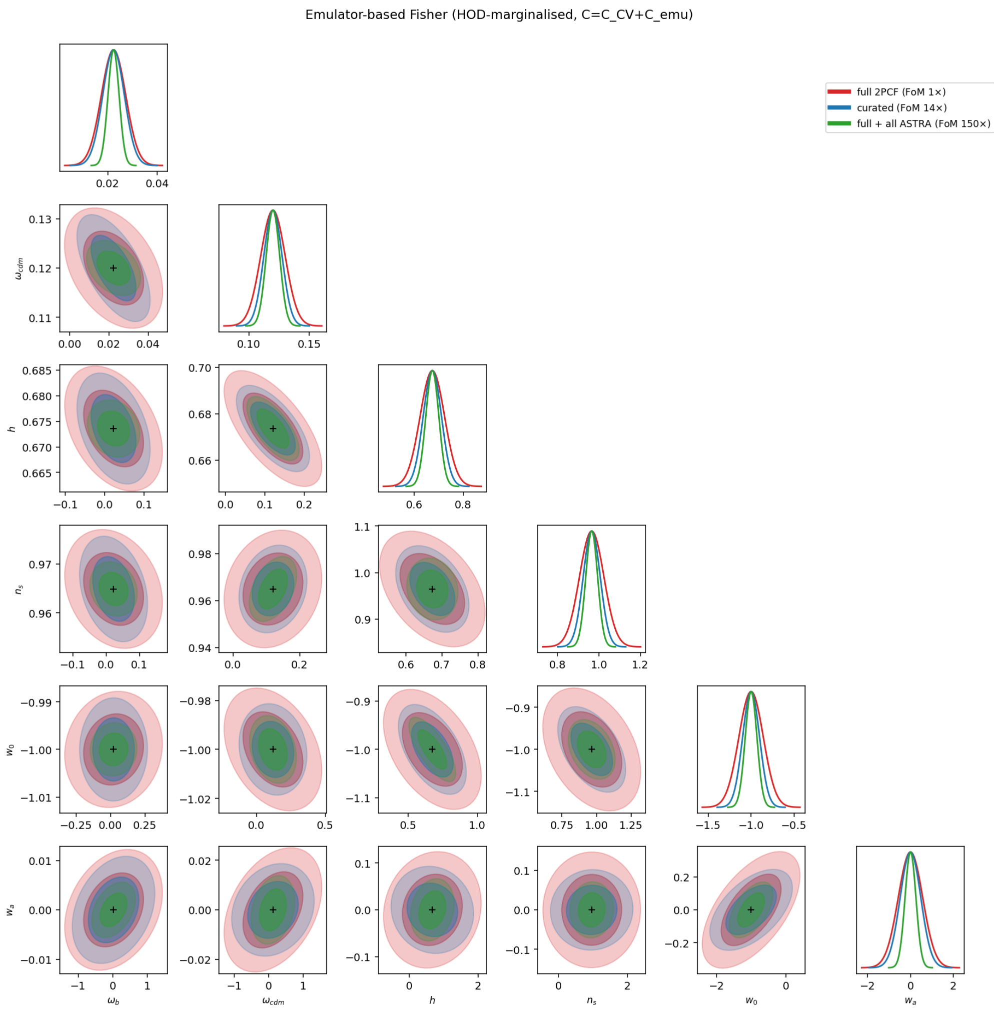

Figure 1: HOD-marginalised corner for the three vectors (legend carries each FoM). The curated and full+ASTRA ellipses are smaller *and* rotated relative to the 2PCF.

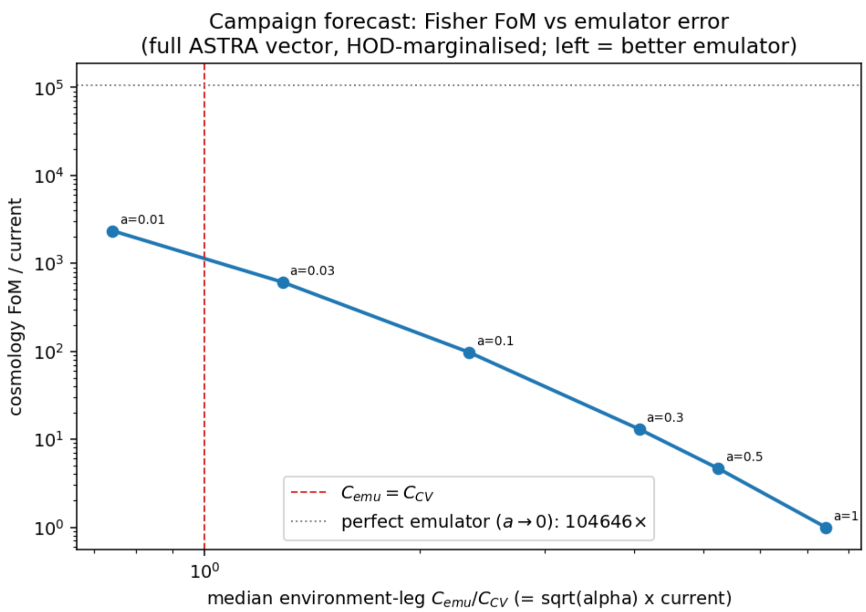

Figure 2: Cosmology FoM vs the median environment-leg $C_{\rm emu}/C_{\rm CV}$ (full ASTRA vector, HOD-marginalised). Left = better emulator; the dashed line marks $C_{\rm emu} = C_{\rm CV}$.